SeoulMMOD: A Large-Scale Multimodal Origin–Destination Flow Benchmark

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Abstract

Origin-destination (OD) flow forecasting supports urban mobility decisions by predicting the volume and patterns of human movement across a city. Most existing public OD benchmarks provide only a partial view of citywide mobility, covering a single transport service or at most two modes. With partial mode coverage, observed demand changes are difficult to separate into shifts across transport modes and changes in total travel. We introduce SeoulMMOD, a large-scale public benchmark for citywide multimodal mobility in Seoul. SeoulMMOD provides three years of hourly OD flows estimating total mobility for 6 urban travel modes across 25 districts and 426 sub-districts, together with travel-time, travel-distance, calendar, rainfall, and point-of-interest (POI) signals. Using SeoulMMOD, we benchmark 14 forecasting baselines under a common protocol and analyse the effects of spatial scaling, multi-mode versus single-mode inputs, and cross-year generalization. Results show that several spatio-temporal models that work at district level do not scale to sub-district OD forecasting. Multi-mode inputs improve single-mode forecasts, but predicting all modes with one model increases error. Together, the dataset and benchmark establish a reproducible testbed for multi-mode, multi-year OD forecasting at the city scale.

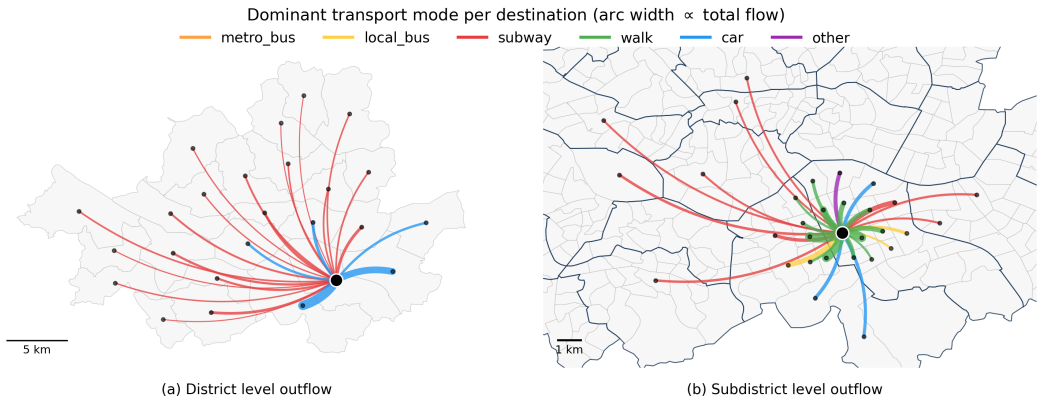
1 Introduction

Large cities move people across rail, bus, road, walking, and shared-mobility systems every day. These movements affect congestion, crowding, and service demand across the city. Urban transport operators need forecasts of how many trips will occur from one zone to another [25]. An **origin-destination (OD) flow** counts trips from an origin zone to a destination zone within a fixed time window. Collecting these flows over all zone pairs gives an OD matrix, a standard representation of citywide mobility. **OD flow forecasting** predicts future trip volumes between zone pairs from recent observations and feeds directly into operational decisions such as timetable optimization on congested rail corridors [24], at-stop control of bus networks to reduce bunching and overcrowding [7], and demand-aware rebalancing of shared-mobility fleets [27].

OD forecasting relies heavily on the availability of OD data. OD flow is difficult to observe directly because trips are distributed across many transport modes and operators, and capturing every trip requires combining their separate records. OD matrices have traditionally been built in two ways. Household travel surveys [5] directly collect reported trips but have limited sample sizes and low update frequencies. Roadside traffic counts [4] infer trip flows from aggregated sensor data, but sparse coverage and non-unique solutions limit these esti-

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(a) District level outflow (b) Subdistrict level outflow

Figure 1: Annual OD outflow in SeoulMMOD (2024) from the same Seoul origin at district and sub-district levels. Arc color indicates the destination's dominant transport mode.

mates. Both produce static OD matrices and cannot capture how flows shift hour to hour or day to day.

Recent OD datasets are built from several operational data sources, including smart-card and fare-gate logs from transit agencies [10], trip records from ride-hailing, taxi, and bike-sharing platforms [37], [40], GPS traces from taxi fleets [41], and mobile-phone signalling data from telecommunication carriers [1], [6], [11], [35]. These records include trip-level timestamps, which make it possible to construct OD aggregates at hourly or finer resolution.

Many public OD datasets built from these sources provide only a partial view of citywide mobility. Existing benchmarks usually cover a single transport service or mode, and multi-mode datasets in the forecasting literature cover at most two. Modes can act as substitutes when travelers switch between alternatives, or as complements when travelers combine them through transfers, such as bus-to-subway connections [42]. With partial mode coverage, observed demand changes are difficult to separate into shifts across modes and changes in total travel. For instance, after a rail service interruption, lower subway demand may reflect switching to buses or taxis, or cancelled trips, each calling for a different operational response. These dependencies motivate forecasting models that use information across modes. To our knowledge, no public OD benchmark provides hourly citywide flows broken down by all major urban transport modes.

To address these gaps, we introduce SeoulMMOD, an hourly citywide OD benchmark covering all major urban transport modes.

Contributions.

1. We release SeoulMMOD, the first public multi-mode time-series city-scale OD forecasting benchmark. It covers six transport modes over three years (2023–2025) at hourly resolution and two spatial levels (district and sub-district), and includes aligned travel-time, travel-distance, holiday, rainfall, and point-of-interest (POI) signals for broader mobility modeling.

2. We benchmark 14 forecasting baselines on SeoulMMOD, providing reference performance and showing that several spatio-temporal models that work at district level do not scale to sub-district OD forecasting.

3. We evaluate multi-mode versus single-mode inputs and cross-year generalization under a unified multi-mode, multi-year protocol. Multi-mode inputs improve single-mode forecasts, but predicting all modes with one model increases error. This identifies a need for models that use multi-mode information while preserving mode-specific demand patterns.

2 Related Work

2.1 OD Forecasting

OD forecasting is a high-dimensional spatio-temporal forecasting problem, where each time step is represented by a $Z \times Z$ OD matrix. The future flow of each OD pair shares temporal patterns with related pairs, such as nearby origins or competing destinations. Deep learning models have been used to capture temporal dynamics and dependencies among OD pairs. Toqué et al. [33] and Liu et al. [17] use LSTMs for public-transport and taxi OD forecasting. To model spatial interactions in OD matrices, Wang et al. [37] use grid embeddings, Jiang et al. [12] use two-dimensional graph convolutions, and Huang et al. [9] use attention mechanisms across OD pairs. Liu et al. [19] use an MLP-based architecture with separate modules for temporal mixing, origin/destination-view spatial mixing, and trend modeling.

Recent work predicts mobility demand across multiple transport modes jointly. Zhang et al. [42] jointly predict subway-bus and subway-bike OD demand and report improvements over single-mode variants. At the zone level rather than the OD-pair level, Liang et al. [15] jointly predict subway and ride-hailing inflow/outflow demand with multi-relational GNNs, and Ke et al. [13] jointly predict solo and shared ride-hailing demand with multi-task multi-graph learning. Together, these studies show that multi-mode inputs can improve forecasting, but existing studies typically use data that span less than one year and cover at most two transport modes.

2.2 Existing OD Datasets

Public OD datasets fall into two categories. Static OD datasets provide a single time-invariant OD matrix per zone pair, often representing commuting flows between residential and workplace regions. For example, LEHD Origin-Destination Employment Statistics (LODES) dataset[2] provides annual home-to-work commuting counts that have been used to build commuting OD datasets at U.S. county and New York City census-tract scales [21], [26]. A single annual snapshot cannot represent hourly demand swings, weekday-weekend differences, or seasonal effects needed for operational forecasting.

Dynamic OD datasets provide OD matrices that change over time, often aggregated from trip records, smart-card taps, or other time-stamped mobility records at hourly or finer resolution. Forecasting studies have used ride-hailing OD matrices from ride-hailing platforms such as UCAR and Didi [37], smart-card metro OD in Chinese cities [18], and subway-bus or subway-bike OD data [42]. Ma et al. [23] aggregate public NYC taxi and bike trip records into regional inflow, outflow, and OD matrices.

Existing dynamic OD datasets share several limitations. First, public datasets typically reflect a single operator, and multi-mode forecasting studies cover at most two modes. In some studies that combine records across operators, modes are not aggregated to a common spatial unit, requiring additional alignment. Second, existing public benchmarks remain small relative to forecasting all OD pairs across a city. Success on a station network or coarse grid does not imply that a method scales to finer citywide administrative zones. Third, temporal spans of weeks to months make it hard to evaluate seasonal patterns or cross-year transfer. Table 1 summarises these datasets alongside SeoulMMOD.

3 SeoulMMOD Dataset

3.1 Data Source and Collection

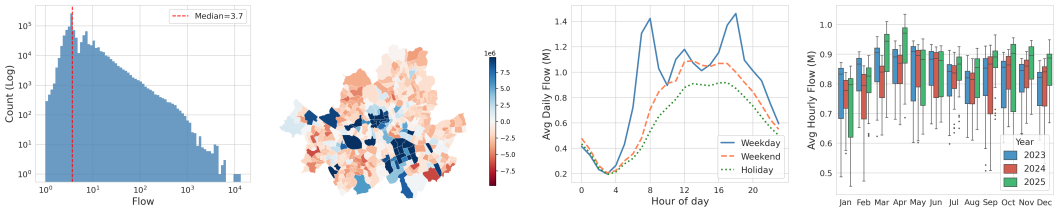
We build SeoulMMOD from publicly available "Seoul Metropolitan Area Living Mobility Data," jointly produced by the Seoul Metropolitan Government and KT (a major South Korean mobile carrier) and distributed through Seoul's open-data portal[3]. According to the public source documentation, the records are derived from KT LTE/5G cell-tower signalling logs, a well-established source for human mobility analysis [8] and OD-demand estimation

[2] https://lehd.ces.census.gov/data/

[3] https://data.seoul.go.kr

Table 1: Related dataset comparison. *Trip attrs* lists per-record auxiliary attributes (average travel time / distance). *Context* lists bundled signals (holiday, point-of-interest, weather). "+" denotes separate node sets for different modes.

Dataset	Coverage	Modes	# Nodes	# OD pairs	Trip attrs	Context	Public
Zone-level forecasting							
Liang et al. [15]	2018-03 – 2018-08	subway, ride-hailing	136 + 63	—	×	×	×
Ke et al. [13]	2018-01 – 2018-12	solo ride-hailing, shared ride-hailing	63	—	×	×	×
Yao et al. [40]	2015-01 – 2015-03 / 2016-07 – 2016-08	bike, taxi	200	—	×	×	✓
OD-level forecasting							
Zhang et al. [42] (BJ)	2016-02 – 2016-04	subway, bus	276 + 2,320	28,802	×	×	×
Zhang et al. [42] (NYC)	2024 (84d, sampled)	subway, bike	121 + 1,081	48,474	×	×	×
UCAR [37]	2016-08 (31d)	ride-hailing	400	160,000	×	Wthr., POI	×
DIDI [37]	2016-11 (30d)	ride-hailing	225	50,625	×	×	×
Deep-NYC-Taxi-Bike [23]	2019-01 – 2020-12	taxi, bike	69	4,761	×	×	✓
HZMOD [18]	2019-01 (25d)	subway	80	6,400	×	×	✓
SHMOD [18]	2016-07 – 2016-09	subway	288	82,944	×	×	✓
SeoulMMOD (ours)	**2023-01 – 2025-12**	**metro bus, local bus, subway, walk, car, other**	**25 / 426**	**625 / 181,476**	**Avg time, Avg dist**	**Hol., POI, Wthr.**	✓



(a) Flow distribution (b) Net inflow (c) 24-hour profile (d) Monthly trend

Figure 2: Aggregate and temporal characteristics of SeoulMMOD. (a) Distribution of OD-record flow. (b) Morning-rush net inflow, computed as inflow minus outflow during 07:00–09:00. (c) Hourly profile by day type. (d) Monthly trend across years.

validated against travel surveys [32]. The original source aggregates these records into administrative-zone OD counts by mode and time, with population-scaling corrections that aim to reduce estimation error from relying on one carrier's subscriber base. Each record reports an estimated total trip volume together with the average travel distance and average travel time for that OD pair, mode, and time bin, auxiliary attributes rarely included in previous public OD benchmarks. Appendix D summarises the original source generation procedure described in the public source documentation and documents our preprocessing steps.

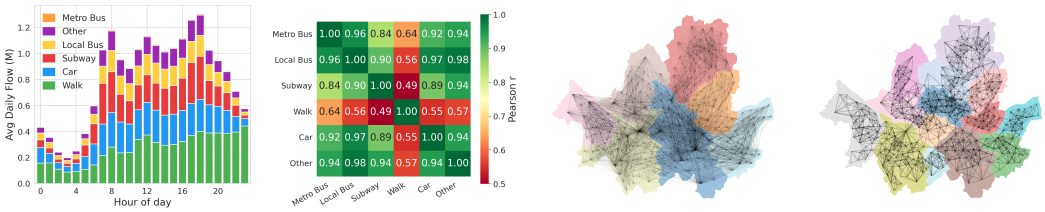
To construct SeoulMMOD, we filter the original source records to OD pairs whose origin and destination both lie within Seoul. We keep the six intra-urban modes (metro bus, local bus, subway, walk, car, other) and drop the three intercity modes (air, train, express bus). We reconcile administrative-code changes during 2023–2025 so every sub-district has a stable identity, and unify the source time bins to a uniform 1 h grid. After these steps, SeoulMMOD spans 2023-01-01 through 2025-12-31, covering 1,096 days and 26,304 hourly time-steps at two spatial levels, 25 districts and 426 sub-districts within Seoul.[4]

3.2 Dataset Analysis

SeoulMMOD shows uneven spatial demand, recurrent temporal cycles, and mode-specific network structure.

Spatio-temporal dynamics. SeoulMMOD varies across space, time, and transport modes. OD flow follows a heavy-tailed distribution (Figure 2a), consistent with observed human mobility patterns in cities [8]. The spatial pattern is uneven. During morning rush hour, net inflow separates destination-heavy sub-districts from origin-heavy sub-districts

[4]The original public data are distributed under the Korea Open Government License (KOGL) Type 1, which permits unrestricted use, redistribution, modification, and commercial use subject only to source attribution. See `https://www.kogl.or.kr/info/licenseTypeEn.do`.



(a) Mode composition (b) Temporal correlation (c) Car backbone (d) Walk backbone

Figure 3: Multi-mode structural diversity. (a) Mode composition over the day. (b) Pairwise Pearson correlation of hourly total flow between modes. (c, d) Disparity-filtered backbone networks and Louvain communities for car and walk, with edge opacity and width proportional to flow. Additional distributional and community-similarity views are shown in Appendix A.

(Figure 2b). Figure 2c shows how flow changes over the day. Peaks occur around 8:00 and 18:00 due to commuting, and disappear on holidays. Total flows vary by month and year (Figure 2d), showing longer-term shifts beyond the daily commuting cycle.

Multi-mode structural diversity. The six modes show different daily activity profiles (Figure 3a) and different temporal correlations (Figure 3b). For instance, Walk becomes dominant late at night, when other transport modes are unavailable. Additional distributional views in Appendix A show that all modes are heavy-tailed, but differ in OD-record flow scale and distance. To examine whether these differences extend to spatial organisation, we construct a network from each mode's OD matrix, apply the disparity filter to extract the backbone [28], and run Louvain community detection [3]. Figures 3c and 3d show the results. Car trips show long-range connections spanning the city, which leads to low modularity and few large clusters (Q=0.584, 7 clusters). In contrast, walk trips show short-range connections that form many localised clusters (Q=0.852, 14 clusters). The modes differ in spatial community structure, indicating that each mode connects Seoul's zones in a different way.

3.3 Auxiliary Data

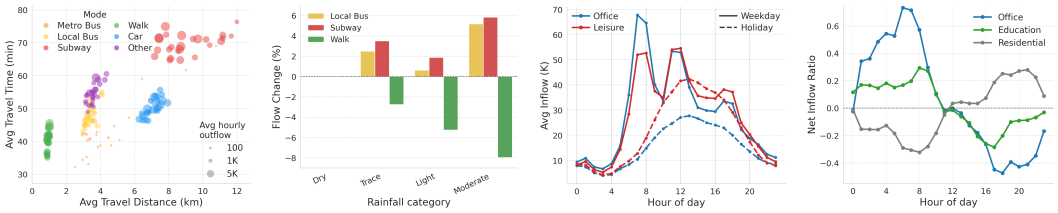
SeoulMMOD provides auxiliary data rarely included in previous public OD benchmarks, including trip attributes and contextual signals. Figure 4 shows representative descriptive views of these fields, and Appendix B provides further descriptive and predictive analyses.

Trip attributes. Each OD record includes average travel time and average travel distance for the corresponding OD pair, mode, and hour. Figure 4(a) relates these attributes to average hourly outflow, showing that each mode concentrates its larger flows in different travel-distance and travel-time ranges.

Calendar. A daily table aligned with the OD time index records day of week, weekend/weekday flag, Korean public-holiday flag, and holiday name. The table covers all 1,096 days of the dataset. Figure 4(c) shows how weekday and holiday inflow profiles differ. Office destinations have a sharp weekday morning peak that is much weaker on holidays, while leisure destinations shift toward broader daytime activity.

Rainfall. Hourly precipitation totals at district level for the full 2023–2025 window, sourced from the Seoul Metropolitan rainfall observation network and distributed through the Seoul Open Data Plaza. Figure 4(b) shows per-mode flow changes under rainfall categories, with lower walk flow and higher local-bus and subway flow on rainy hours.

POI categories. Per-sub-district counts of points of interest from the Seoul Open Data Plaza commercial-district POI release. We use the 2025 snapshot, and the source is updated once per year. These counts cover 20 categories, spanning transport infrastructure, commerce and leisure, public and health services, and education. Figure 4(d) compares weekday net-inflow roles for selected office, education, and low-POI residential examples. Office and



(a) Mode attributes (b) Rainfall by mode (c) Calendar shift (d) POI examples

Figure 4: Trip attributes and contextual information in SeoulMMOD. (a) District-level travel distance and travel time, with marker size showing average hourly outflow. (b) Per-mode hourly flow change under rainfall categories, relative to dry hours. (c) Hourly inflow for office and leisure destination examples on weekdays and holidays. (d) Weekday net-inflow ratio for selected office, education, and low-POI residential examples.

education examples receive net inflows in the morning and shift toward net outflows later in the day, while the residential example follows the reverse pattern.

4 Experiments

We use SeoulMMOD to investigate how existing forecasting models behave across multiple transport modes, spatial scales, and temporal horizons. These experiments cover forecasting accuracy, computational cost, multi-mode versus single-mode inputs, and cross-year generalization.

4.1 Setup

Task. We adopt the standard multi-horizon OD forecasting setting. Given the past T hours of OD flow $\mathbf{X}_{t-T+1:t} \in \mathbb{R}^{T\times P\times M}$ across $P{=}Z^2$ OD pairs from Z zones and $M = 6$ transport modes, the model predicts the next H hours $\hat{\mathbf{X}}_{t+1:t+H}$. We evaluate three forecasting horizons $H \in \{6, 12, 24\}$ with input length matching the horizon ($T = H$). Splits are chronological with a 70/10/20 train/validation/test ratio over the 2024 calendar year. To evaluate cross-year generalization, Section 4.5 evaluates models trained on the 2024 training split on 2025 data with the same input/output configuration.

Baselines. We benchmark 14 models in total. The first two are non-learned baselines. Historical Average (HA) predicts the mean value for each (hour-of-week, OD pair, mode) combination, computed on the training portion. Latest Value (LV), also known as persistence, repeats the most recent observed value across the forecast horizon.

The remaining 12 are learned baselines, each using one model with all six modes as both inputs and outputs in the main benchmark. STAEformer [16], STID [30], AGCRN [2], Graph WaveNet [39], and MTGNN [38] operate on the flattened OD-pair representation. iTransformer [20], SOFTS [22], MTSMixer [14], and TimeMixer [36] are trained on one OD pair at a time and treat the six modes as interacting variates. MPGCN [31] and ODCRN [12] preserve the $Z{\times}Z$ OD matrix structure and apply graph convolution along the origin and destination axes. ODMixer [19] adopts a factorized MLP design for fine-grained OD forecasting.

Implementation. All learned baselines share the same chronological splits, forecasting horizons, early-stopping budget, and evaluation metrics. They are run through BasicTS-compatible configurations and runners [29]. For MPGCN, ODCRN, and ODMixer, model-specific adapters preserve the $Z{\times}Z$ OD-matrix layout while writing metrics in the same format. We report two types of failures. Out-of-memory (OOM) indicates that a model exceeds the GPU memory budget, while time-exceeded (TE) means its per-epoch wall-clock time exceeds one hour. All experiments are conducted on NVIDIA RTX PRO 6000 (97 GB) GPUs. Model-specific wrappers and other implementation details are provided in Appendix C.

Metrics. We report three metrics in estimated trip-count units. Mean absolute error (MAE) serves as the headline metric throughout the paper. Root mean squared error (RMSE) additionally penalises large individual prediction errors. Weighted mean absolute percentage error, wMAPE $= \sum |\hat{y} - y| / \sum |y|$, is commonly reported in OD forecasting work [19], [42] and provides a percentage-scale measure that avoids per-observation division by zero.

4.2 Multi-Horizon Performance

Table 2 reports overall MAE, RMSE, and wMAPE at 6h, 12h, and 24h. The rightmost columns report 24h computational cost.

At the district level, spatial-interaction models achieve the lowest errors. STAEformer, which applies attention over the flattened OD-pair representation, reports the lowest overall MAE at all three horizons. GNN-based models such as AGCRN, GWNet, and MTGNN generally follow STAEformer. Channel-decomposed transformers exhibit larger errors at 6h and 12h, but move closer to the GNN-based models at 24h. Among non-learned baselines, HA remains competitive with an overall MAE only 6.1% behind STAEformer at 24h, while LV is uncompetitive at every horizon, with an overall MAE two to three times that of HA.

Across the three horizons, the learned baselines separate into three patterns. STAEformer and STID are nearly horizon-invariant, with district MAE varying little from 6h to 24h. Several spatio-temporal models without temporal embeddings degrade at longer horizons, including AGCRN, GWNet, MTGNN, MPGCN, and ODMixer. By contrast, iTransformer, SOFTS, MTSMixer, and TimeMixer drop from an average MAE of 59.8 at 6h to 28.8 at 24h. In our configurations, STAEformer and STID include explicit time-of-day and day-of-week embeddings, while the pair-wise time-series models use only flow histories. HA's strong performance with just an hour-of-week lookup suggests that recurring daily and weekly rhythms dominate the OD signal. Temporal embeddings encode these rhythms directly, while pair-wise models recover them once the input window covers a full daily cycle.

At the sub-district level, the horizon comparison is limited to the few baselines that complete under the common evaluation protocol. Among completed learned models, ODMixer gives the strongest short-horizon results at 6h and 12h, while STID is strongest at 24h.

For the per-mode breakdown (Table 3), we focus on 24h because it exposes the largest spread between architectures and is the most operationally meaningful for day-ahead planning. HA leads on metro bus, subway, and car, whose hourly profiles are dominated by the weekly cycle. STAEformer leads on local bus, walk, and other. Its 19.6% reduction in walk MAE alone accounts for most of its 6.1% lower average MAE relative to HA. On these modes, learned models capture complex patterns that the weekly cycle alone misses.

4.3 Scalability and Computational Cost

Forecasting OD demand at the sub-district level expands the prediction target by two orders of magnitude, from 625 to 181,476 OD pairs. We therefore test whether existing models remain computationally feasible at this scale. The rightmost columns of Table 2 report per-baseline parameter count, batch size, per-epoch wall-clock time, and estimated 100-epoch wall-clock time at the 24h horizon for both spatial levels.

At the district level, per-baseline training cost spans nearly two orders of magnitude. STAEformer achieves the lowest 24h MAE among learned models, but its 2.8M parameters and over two hours of training time make it inefficient compared to other baselines. STID, the smallest spatio-temporal baseline, completes training within minutes, with 15.3% higher 24h MAE than STAEformer, illustrating the trade-off between accuracy and cost.

At the sub-district level, the limiting factor depends on the architecture. Models that explicitly capture spatial interaction across OD pairs exceed the available memory, whereas models that treat each OD pair as an independent sequence fit in memory but exceed the per-epoch time budget by an order of magnitude. MPGCN and ODCRN, for example, keep the OD data as dense arrays with shape $[B, T, Z, Z, 6]$ and store hidden activations over both origin and destination axes. At Z=426 these activations dominate the small number

Table 2: Multi-horizon performance and computational cost, aggregated across all six modes. Each learned baseline uses all six modes as inputs and outputs. Bold marks the best and underline marks the second-best per (horizon, metric) within each level. Cost columns report parameter count, batch size, per-epoch wall-clock, and 100-epoch wall-clock estimates at the 24h→24h horizon.

	6h→6h			12h→12h			24h→24h			Cost (24h)			
Model	MAE	RMSE	wMAPE	MAE	RMSE	wMAPE	MAE	RMSE	wMAPE	#Params	BS	Train/ep	Total
District scale (*625 OD pairs*)													
HA	22.9	112.2	10.5%	22.9	112.2	10.5%	22.8	111.8	10.5%	—	—	—	—
LV	106.6	566.8	49.0%	144.2	692.3	66.3%	139.0	678.8	63.9%	—	—	—	—
STAEformer	**21.8**	**93.7**	**10.1%**	**22.8**	**100.1**	**10.5%**	**21.4**	**92.9**	**9.8%**	2.8M	8	83s	2.3h
STID	23.4	99.7	10.8%	25.1	106.9	11.5%	24.6	111.7	11.3%	143K	16	3.2s	0.09h
AGCRN	22.7	102.6	10.4%	25.2	111.2	11.6%	30.4	178.9	14.0%	780K	32	21s	0.57h
GWNet	23.8	95.9	10.9%	26.2	106.8	12.1%	27.6	118.8	12.7%	353K	32	31s	0.86h
MTGNN	23.8	101.9	11.0%	26.7	112.4	12.3%	27.4	120.1	12.6%	1.7M	32	14s	0.38h
iTransformer	56.7	253.8	26.1%	38.7	162.8	17.8%	25.3	119.3	11.7%	4.8M	256	160s	4.4h
SOFTS	58.4	266.3	26.9%	42.6	180.6	19.6%	26.5	130.1	12.2%	205K	256	145s	4.0h
MTSMixer	67.8	329.1	31.2%	51.5	243.8	23.7%	31.9	172.7	14.7%	14K	256	118s	3.3h
TimeMixer	56.1	265.7	25.9%	51.3	253.2	23.6%	31.6	173.0	14.5%	16K	256	492s	13.7h
ODCRN	27.8	192.6	12.8%	31.8	226.3	14.7%	30.2	226.6	13.9%	178K	16	233s	6.5h
MPGCN	34.9	232.8	16.1%	40.5	276.7	18.7%	43.0	299.2	19.8%	75K	16	14s	0.39h
ODMixer	31.2	238.7	14.4%	37.8	269.2	17.4%	51.9	378.0	23.9%	206K	16	6.8s	0.19h
Sub-district scale (*181,476 OD pairs*); *trained for up to 100 epochs with early-stopping* (*patience 10*)													
HA	0.475	**2.81**	63.5%	0.475	**2.81**	63.5%	0.475	**2.80**	63.5%	—	—	—	—
LV	0.742	9.41	99.1%	0.833	11.0	111.2%	0.820	10.9	109.6%	—	—	—	—
STAEformer	—	—	—	—	—	—	—	—	—	350.0M	8	OOM	OOM
STID	0.434	5.20	58.4%	0.448	5.79	60.2%	**0.442**	4.81	**59.2%**	5.9M	16	1820s	50.5h
AGCRN	—	—	—	—	—	—	—	—	—	2.6M	32	OOM	OOM
GWNet	—	—	—	—	—	—	—	—	—	4.0M	32	OOM	OOM
MTGNN	—	—	—	—	—	—	—	—	—	432.8M	32	OOM	OOM
iTransformer	—	—	—	—	—	—	—	—	—	4.8M	256	TE	TE
SOFTS	—	—	—	—	—	—	—	—	—	205K	256	TE	TE
MTSMixer	—	—	—	—	—	—	—	—	—	14K	256	TE	TE
TimeMixer	—	—	—	—	—	—	—	—	—	16K	256	TE	TE
ODCRN	—	—	—	—	—	—	—	—	—	540K	16	OOM	OOM
MPGCN	—	—	—	—	—	—	—	—	—	75K	16	OOM	OOM
ODMixer	**0.423**	5.21	**56.6%**	**0.434**	5.54	**58.1%**	0.485	6.96	64.9%	46.7M	16	540s	15.0h

Table 3: Per-mode MAE and RMSE at the 24h→24h horizon, district level. Bold marks the best and underline marks the second-best per (mode, metric).

	metro bus		local bus		subway		walk		car		other		Avg	
Model	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE
HA	**2.00**	**7.85**	18.7	72.4	**32.2**	73.2	35.2	233.4	**30.5**	74.7	18.1	64.9	22.8	111.8
LV	6.11	34.3	104.9	417.8	231.1	466.1	232.1	1429.7	160.0	436.3	99.8	371.0	139.0	678.8
STAEformer	2.30	11.2	**16.5**	**55.4**	33.0	**72.0**	**28.3**	**188.9**	31.1	**69.4**	**17.0**	**54.3**	**21.4**	**92.9**
STID	2.94	13.9	19.4	65.2	37.4	82.7	33.1	231.7	35.3	78.9	19.6	60.9	24.6	111.7
AGCRN	2.83	13.1	21.7	79.0	45.0	103.9	53.0	401.8	38.0	89.0	21.7	73.4	30.4	178.9
GWNet	2.66	12.3	21.2	70.8	41.7	84.8	41.5	248.1	37.4	81.1	20.9	64.4	27.6	118.8
MTGNN	3.00	12.9	20.8	70.5	44.6	96.3	37.9	247.4	37.1	81.9	21.0	65.0	27.4	120.1
iTransformer	2.07	9.03	18.7	69.4	44.5	112.6	31.9	237.9	35.2	82.0	19.6	67.7	25.3	119.3
SOFTS	2.16	9.96	19.5	74.1	47.1	125.6	32.9	259.4	36.8	87.3	20.6	72.6	26.5	130.1
MTSMixer	3.53	14.6	23.1	87.7	57.5	154.7	42.9	361.1	41.2	98.6	23.6	83.6	31.9	172.7
TimeMixer	2.73	13.0	22.4	90.3	57.8	160.8	41.4	356.0	41.5	104.5	23.6	87.7	31.6	173.0
ODCRN	2.12	12.0	20.9	82.4	37.6	83.1	58.2	523.6	34.6	85.4	20.3	79.0	28.9	224.2
MPGCN	2.55	11.7	25.1	90.5	56.8	126.7	57.4	388.1	45.0	105.2	24.4	82.6	35.2	179.3
ODMixer	5.49	26.9	29.9	122.8	64.5	147.3	79.2	610.0	52.6	141.3	31.5	120.5	43.9	272.0

of stored weights. Pair-independent transformers avoid dense $Z \times Z$ hidden states, but they train over 181,476 OD-pair sequences, which drives the per-epoch wall-clock past the budget.

4.4 Multi-Mode vs Single-Mode Inputs

A multi-mode forecasting model can use the target mode's history alone or the histories of all six modes. We compare these inputs for STID and STAEformer at the district 24h

Table 4: Multi-mode input/output comparison: average MAE across six modes, district 24h horizon. Single-target settings train six models (one per target). Bold marks the lowest MAE; — denotes not evaluated.

Flow inputs	Flow outputs	Models	STID	STAEformer
Target mode	Target mode	6	20.338	20.205
All six modes	Target mode	6	**20.241**	**19.494**
All six modes	All six modes	1	24.628	21.367
Other five modes	Target mode	6	22.329	—

Table 5: Cross-year generalization at the district 24h horizon (2024 training split, full 2025 evaluation), broken down by season. Bold marks the easiest season per model. Δ% is the MAE change from the 2024 test split to the full 2025 calendar.

Model	2024 test	2025 overall	Spring	Summer	Fall	Winter	Δ
STAEformer	21.37	25.83	28.29	**23.70**	25.75	25.56	+20.89%
AGCRN	30.37	35.77	37.33	**32.84**	36.93	36.00	+17.78%
GWNet	27.55	33.47	35.21	**30.60**	33.84	34.29	+21.49%
MTGNN	27.40	32.35	34.63	**29.95**	31.93	32.89	+18.07%
iTransformer	25.33	28.08	29.87	**26.65**	28.35	27.42	+10.86%

horizon, training one model per target mode in both settings (Table 4). Adding the other five modes' histories reduces average MAE by 0.48% for STID and 3.52% for STAEformer.

We also evaluate one model that takes all six modes as inputs and predicts all six as outputs. Relative to all-six-to-one, this increases MAE by 21.7% and 9.6%, respectively. In our implementation, this setting uses the six modes as input and output channels but does not explicitly model dependencies between modes. A single shared representation must predict targets with different scales, sparsity patterns, temporal profiles, and spatial structures. This suggests that future multi-mode models need mechanisms for mode-specific patterns and interactions between modes. The comparison also changes total model capacity; per-mode results and single-run limitations are provided in Appendix C.4.

4.5 Cross-Year Generalization

We evaluate out-of-distribution (OOD) generalization across years to assess robustness to changes in urban mobility patterns [34]. We evaluate five architectures (STAEformer, AGCRN, Graph WaveNet, MTGNN, and iTransformer) on the full 2025 calendar without retraining, using checkpoints trained on the 2024 training split and selected by validation MAE. Table 5 reports overall and per-season MAE at the district 24h horizon.

All five models have higher MAE in cross-year evaluation, with iTransformer showing the smallest increase. The seasonal ordering is consistent across architectures, with spring being the hardest and summer the easiest.

Contextual extensions. We explore how three contextual signals released with the dataset (holiday calendar, hourly rainfall, per-district POI counts) change forecasting accuracy. Each of the three signals lowers 24h MAE by 0.49 to 1.39 when added to STID, but their effects differ at shorter horizons: the holiday embedding raises MAE at 6h and 12h, rainfall raises MAE at 6h but lowers it at 12h, and POI counts stay neutral or slightly helpful at every horizon. Appendix B.1 gives the full table and setup.

5 Conclusion

We introduced SeoulMMOD, a city-scale multi-mode OD forecasting benchmark for Seoul that combines six-mode hourly flows with trip attributes and contextual information. This benchmark highlights two central challenges for current OD forecasting models in city-scale

multi-mode settings. Spatio-temporal models that perform well at district level do not scale to sub-district forecasting, while the input ablations show that multi-mode inputs improve single-mode forecasts but lose these gains when one model predicts all modes. Future model design should scale to all origin-destination pairs at fine spatial resolution and use multi-mode information without sacrificing mode-specific patterns. Substitution and transfer connections offer transport-informed directions for modeling these dependencies. SeoulMMOD provides a reproducible testbed for developing and evaluating such models.

Reproducibility statement

All learned baselines are run through the same BasicTS-style training and evaluation pipeline [29]. The project repository releases the dataset builders, model configurations, BasicTS-compatible adapters, and training and evaluation scripts used in the benchmark. Appendix C documents the benchmark protocol and model-specific implementation details. The dataset release provides the OD flow parquet files and metadata needed to build the benchmark tensors used for Table 2.

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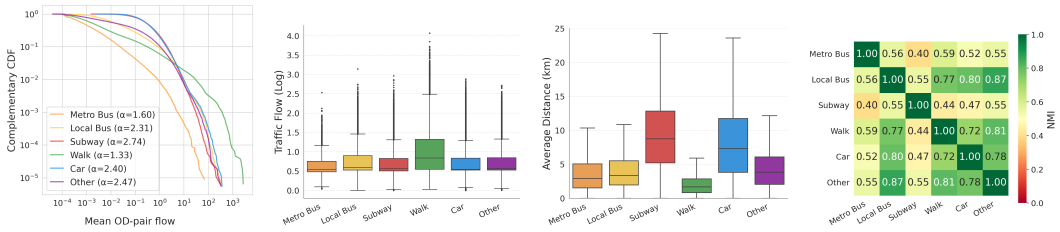
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(a) Heavy-tailed CCDF (b) Flow by mode (c) Distance by mode (d) Community NMI

Figure 5: Additional descriptive views of multi-mode OD structure. (a) Complementary cumulative distribution of OD-pair flows per mode. (b) Distribution of OD-record flow by mode. (c) Distribution of average OD-record distance by mode. (d) Normalised mutual information between per-mode community assignments.

Table 6: Mode share (%) and total daily trip volume by day type for 2024 at the district scale. *Holiday* counts Korean public holidays. *Weekend* excludes days flagged as holidays. Mode shares sum to 100% per row.

Day type	n days	Trips/day	metro bus	local bus	subway	walk	car	other
Weekday	246	20.64 M	0.61	12.48	21.73	33.06	20.66	11.45
Weekend	101	16.87 M	0.59	12.80	17.63	35.06	22.80	11.12
Holiday	19	14.71 M	0.59	12.58	16.24	36.35	23.61	10.63
Change vs weekday (*pp*)			metro bus	local bus	subway	walk	car	other
Weekend	—	−18%	−0.02	+0.32	−4.10	+1.99	+2.14	−0.33
Holiday	—	−29%	−0.02	+0.10	−5.50	+3.29	+2.95	−0.82

A Additional Dataset Analysis

Figure 5 complements Section 3.2 with four descriptive views of the OD data. Mode-specific flow distributions are heavy-tailed, and trip volumes and distances vary by mode, suggesting different hub sizes and interaction ranges per mode. We examine these differences at the network level by comparing Louvain community partitions across modes with normalised mutual information (NMI). Figure 5d shows generally low NMI values, suggesting that spatial community assignments differ across modes.

B Contextual Signal Analysis

SeoulMMOD bundles three contextual signals (Section 3.3) to support context-aware extensions without re-collecting data. We describe each signal (Calendar, Rainfall, POI) and then evaluate predictive utility and error stratification in Sections B.1 and B.2.

Calendar. Table 6 compares total daily trip volume and per-mode share across weekdays, non-holiday weekends, and Korean public holidays in 2024. Total demand drops 18% on weekends and 29% on holidays. The subway share falls by 4.1 pp on weekends and 5.5 pp on holidays. Walking (+3.3 pp on holidays) and car (+3.0 pp) account for larger shares of the smaller holiday total. Their absolute volumes still fall. These aggregate changes show that the mode mix varies with day type, but do not identify individual mode substitutions or trip purposes.

Rainfall. Table 7 reports origin district-hour mode shares by rainfall intensity. Trips on dry district-hours are split 20.5% subway, 33.7% walk, 21.3% car, with the remainder in bus and other modes. Heavy-rain cells (>10 mm/hour) account for 0.30% of district-hours. In these cells, walk share drops to 30.6% and subway share rises to 22.4%. Rainfall shifts mode shares by a few percentage points, so we release it as an auxiliary feature rather than the main target.

Table 7: Origin district-hour mode share (%) by rainfall intensity, district scale, 2024. Mode shares sum to 100% per row.

Condition	n cells	Share	metro bus	local bus	subway	walk	car	other
Dry (0 mm)	206,895	94.21%	0.60	12.55	20.49	33.74	21.30	11.33
Trace (<0.5 mm)	3,012	1.37%	0.67	12.38	21.22	32.44	21.75	11.54
Light (0.5–2 mm)	5,741	2.61%	0.63	12.74	21.09	33.13	20.98	11.43
Moderate (2–10 mm)	3,283	1.49%	0.62	13.04	20.88	32.43	21.44	11.58
Heavy (>10 mm)	669	0.30%	0.62	13.42	22.42	30.60	21.06	11.87

Table 8: POI count features grouped into four broad groups for presentation.

Group	Categories
Transport infrastructure	airport, rail station, bus terminal, subway station, bus stop
Commerce and leisure	attraction, bank, department store, supermarket, theater, hotel
Public and health services	government office, general hospital, clinic, pharmacy
Education	kindergarten, elementary school, middle school, high school, university

POI categories. Per-sub-district POI counts cover 20 categories, which we group into four broad groups for presentation. Table 8 lists the categories in each group. To check whether these counts predict the mode mix at each destination, we correlate each category's count with the share of arrivals using each mode in 2024 (Table 9). For each (POI, mode) pair, we report raw Pearson correlation, Spearman rank correlation, Pearson correlation on POI density, and partial Pearson correlation controlling for sub-district area. FDR-adjusted q-values are computed across all 120 POI–mode pairs, and the table reports entries with raw $|r|\geq 0.20$ and $q<0.001$.

Subway infrastructure gives the clearest signal. Subway-station count correlates with subway arrival share at $r=+0.458$ and remains stable under the rank, density, and area-controlled variants. Bus-stop count tracks local-bus share, while commerce and leisure categories such as banks and theaters track subway share. Education categories, especially kindergarten, correlate negatively with subway share.

For bus stops and walk share, the raw count correlation is negative ($r=-0.284$), but it changes to $r=+0.184$ under density and weakens to $r=-0.097$ after controlling for sub-district area. Sub-district size explains the raw association because large sub-districts can contain many bus stops while having low walk share due to longer typical trip distances.

B.1 Predictive utility of contextual signals

To evaluate whether contextual signals improve forecasts on top of mode-flow history, we extend STID with three variants and measure the change in test MAE on the district-scale 2024 split. STID's embedding-and-MLP design lets each variant add a single lookup or projection in isolation, so per-signal effects are easy to compare. *Holiday* adds a 3-state day-type embedding (workday / weekend / holiday) read from the input and prediction windows. *POI* attaches a static per-pair feature built from 13 district-level POI categories. The origin and destination count vectors are concatenated, log-transformed, z-scored, and projected to a 32-d embedding shared across timesteps. *Rainfall* adds two dynamic channels, hourly rainfall at the origin and destination districts from the Seoul AWS network, log1p-transformed and z-scored. Each dynamic channel passes through a separate 1×1 convolution to keep its projection independent from the mode-flow channels.

Setup. We follow the benchmark protocol described in Appendix C. All STID variants share an embed dim of 32, 3 encoder layers, learning rate $2\cdot10^{-3}$, and MultiStepLR with milestones $\{1, 50, 80\}$ ($\gamma=0.5$). We report the test MAE produced by the best validation checkpoint per (variant, horizon).

All three contextual signals lower 24-hour MAE. Holiday lowers MAE by 1.39, POI by 0.49, and Rainfall by 0.49. At shorter horizons the three signals behave differently. Holiday hurts $h=6$ and $h=12$ (+0.40, +0.48 MAE), so it helps only when the prediction window spans

Table 9: Correlation between per-sub-district POI count and destination mode share. We report four correlation variants per (POI, mode) pair — Pearson r on counts, Spearman r_s, Pearson r on per-km^2 density, and partial r controlling for sub-district area. Rows have raw $|r| \geq 0.20$ and BH-adjusted $q < 0.001$.

POI	Mode	r (count)	r_s	r (density)	r (partial | area)
Subway station	subway	+0.458	+0.442	+0.438	+0.475
Bank	subway	+0.356	+0.314	+0.430	+0.369
Theater	subway	+0.345	+0.298	+0.332	+0.347
Subway station	local bus	−0.284	−0.324	−0.256	−0.286
Bus stop	local bus	+0.276	+0.324	+0.318	+0.338
Kindergarten	subway	−0.283	−0.307	−0.257	−0.283
Attraction	subway	+0.277	+0.221	+0.307	+0.307
Gov office	subway	+0.259	+0.190	+0.295	+0.276
Bank	metro bus	+0.239	+0.367	+0.215	+0.222
Pharmacy	subway	+0.239	+0.209	+0.222	+0.245
Bank	local bus	−0.221	−0.233	−0.223	−0.220
Bus stop	walk	−0.284	−0.219	+0.184	−0.097

Table 10: Test MAE on the 2024 district-scale dataset when STID is extended with each contextual signal in isolation. Lower is better, bold marks the best per horizon, and Δ is the change against *STID base*.

Variant	MAE @ 6	MAE @ 12	MAE @ 24	Δ @ 6	Δ @ 12	Δ @ 24
STID base	23.41	25.07	24.63	—	—	—
+ Holiday	23.81	25.55	**23.24**	+0.40	+0.48	−1.39
+ POI	**22.97**	25.35	24.14	−0.44	+0.28	−0.49
+ Rainfall	24.75	**24.37**	24.14	+1.34	−0.70	−0.49

multiple days. Rainfall reverses sign with horizon, going from +1.34 MAE at h=6 to −0.70 at h=12. POI helps at h=6 and h=24 but costs +0.28 MAE at h=12, so no variant improves MAE at every horizon.

We make the variants, configs, and build scripts available so users can reproduce the table or run multi-seed experiments.

B.2 Error stratification by contextual signals

These contextual signals let users diagnose model behavior after training. As one example, we aggregate 24-hour district-scale test errors by destination district and compare them with the district's POI count. Figure 6 shows that districts with more POIs have higher target volume and larger absolute errors for both HA and STAEformer. These districts have lower wMAPE, showing that absolute error and relative error can tell different stories about the same urban contexts.

This POI-error comparison is diagnostic rather than causal. Districts with more POIs concentrate more demand, so the plot should not be read as POIs causing model error. The point is that SeoulMMOD's contextual signals let users inspect where forecasting errors concentrate across urban environments.

C Benchmark Implementation Details

All twelve learned baselines use the same 70/10/20 chronological train/val/test split, a maximum of 100 epochs, Adam optimization, validation and test evaluation after every epoch, and validation MAE for checkpoint selection. Where early stopping is enabled, the patience is 10 epochs. Learning rates, batch sizes, weight decay, gradient clipping, and learning-rate schedules remain model-specific. Each baseline follows its source configuration or framework default, with level-specific changes recorded in the released configs. In the main benchmark, all learned models predict all six modes (M=6) over each forecast horizon

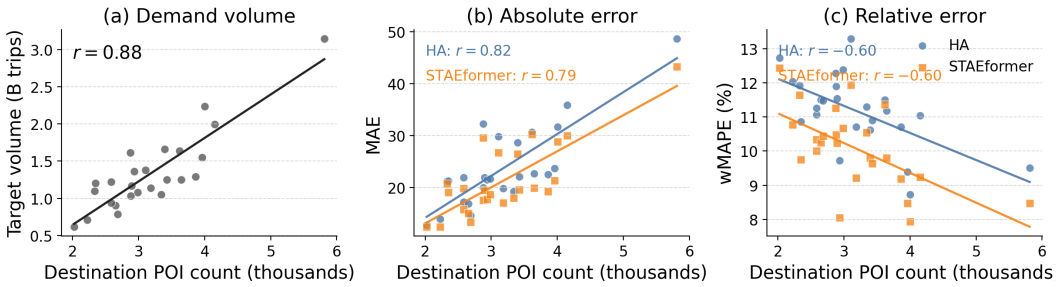


Figure 6: Error stratification by destination-district POI count. (a) Districts with more POIs have larger target volume. (b) Absolute MAE increases with POI count for HA and STAEformer. (c) Relative error decreases with POI count, indicating that high-volume districts can have larger absolute errors but smaller percentage errors. Pearson correlations are shown in each panel.

and report MAE, RMSE, and wMAPE after inverse scaling. Throughout this appendix, Z denotes the number of zones and $P{=}Z^2$ the number of OD pairs.

C.1 Spatio-temporal forecasting baselines

Nine spatio-temporal baselines (STAEformer, STID, AGCRN, Graph WaveNet, MTGNN, iTransformer, SOFTS, MTSMixer, TimeMixer) run within BasicTS [29] with masked-MAE loss. STAEformer, STID, AGCRN, Graph WaveNet, and MTGNN use the BasicTS Z-score scaler, while iTransformer, SOFTS, MTSMixer, and TimeMixer use a global Z-score variant for pair-wise time-series inputs. In all cases, scaling is applied only to the six mode-flow channels; calendar, holiday, distance, time, and rainfall channels are passed through unchanged.

Graph WaveNet and MTGNN expose an `out_dim` argument meant for single-channel multi-step output. We wrap them with an adapter that sets `out_dim`$=H\times6$ and reshapes the flat output to $[B, H, P, 6]$.

C.2 OD flow forecasting baselines

MPGCN, ODCRN, and ODMixer retain the $Z\times Z$ OD layout and their core graph, recurrent, or mixer operations. BasicTS-compatible adapters expose six-mode inputs and outputs under the common benchmark protocol. All three use training-fitted, per-mode standardization of $\log(1+x)$ flows to reduce their dynamic range. MAE is computed after inverse scaling; validation and test predictions are clipped to zero in flow units.

MPGCN. We retain the per-OD LSTM, static and dynamic graph branches, bidimensional graph convolutions, and branch averaging. The single-step head is expanded to $H\times6$ outputs for direct multi-horizon prediction. Its output ReLU is removed because low flows can have negative standardized values. For hourly Seoul data, the released configuration uses a uniform static graph and 24 hour-of-day graphs derived from training-only flows summed over modes. These use cosine similarities between rows (origins) and between columns (destinations), replacing the reference loader's seven-step period and cosine-distance construction. Matrix products and chunked accumulation make graph preprocessing practical at larger spatial scales.

ODCRN. We retain the OD-convolutional recurrent cells, dynamic graph constructor, and autoregressive encoder–decoder. The adapter sets the input/output channels to six, supplies Seoul's static adjacency, and uses the shared scaling and MAE-based training pipeline. Thus the adaptation preserves the recurrent OD dynamics while accommodating multi-mode observations and the benchmark horizons.

Table 11: Per-mode MAE for the multi-mode input/output ablations, district 24h horizon. Each single-target setting uses one model per target mode. Values are reported to three decimal places.

Model	Flow inputs → outputs	Metro bus	Local bus	Subway	Walk	Car	Other	Avg.
STID	Target → target	1.792	15.849	30.976	27.765	29.377	16.269	20.338
	All six → target	1.780	15.822	30.861	27.912	29.035	16.038	20.241
	All six → all six	2.937	19.364	37.427	33.138	35.319	19.580	24.628
	Other five → target	1.864	16.972	33.814	33.926	30.571	16.831	22.329
STAEformer	Target → target	1.815	16.011	29.375	29.798	27.905	16.324	20.205
	All six → target	1.770	15.183	29.371	26.880	27.944	15.814	19.494
	All six → all six	2.299	16.514	32.951	28.330	31.123	16.985	21.367

ODMixer. We retain the Channel Mixer, Multi-view Mixer, and Bidirectional Trend Learner. The input projection receives $T \times 6$ time–mode features per OD pair, and the output projection produces $H \times 6$ values; the spatial axes remain unchanged. The second branch receives the window shifted back by 24 hours and retains its auxiliary MAE loss, preserving daily trend learning while extending the original single-mode, single-step interface. These adaptations support the benchmark task; their individual accuracy effects are not isolated by ablations.

C.3 Parameter-free baselines

HistoricalAverage and LatestValue share the BasicTS preprocessing and runner code with the learned baselines. HistoricalAverage builds a 168-bucket hour-of-week table over the training portion in normalized flow space and maps each test timestamp to its bucket via the dataset's tod/dow channels. LatestValue copies the last input frame across the prediction horizon.

C.4 Multi-mode input and output settings

The ablations in Section 4.4 use the 2024 district split with 24h input and prediction windows. Target-only uses the target mode's flow history; all-six-to-one uses all mode histories to predict one target; all-six-to-all-six uses one model to predict all modes. For STID, other-five-to-one excludes the target's history but still trains with that target's labels, so it does not evaluate transfer to an unseen mode. Its average MAE is 22.329, 9.8% above target-only; other modes' histories do not replace the target's own history in this setting. Time-of-day and day-of-week inputs are retained across settings. We select each checkpoint by validation MAE and report one run per configuration, without a multi-seed significance claim. Table 11 reports each mode and their arithmetic mean; all modes have the same number of evaluation entries.

The input comparison holds the number of models at six while changing the available flow histories. For STID, target-only uses 123,960 parameters per model (743,760 total), and all-six-to-one uses 127,800 per model (766,800 total). The single six-output model uses 143,280 parameters. Consequently, this comparison changes both output sharing and total capacity, and does not establish that either alone causes the error increase. In STID, the smallest-volume mode, metro bus, has 65.1% higher MAE when predicting all six modes than under all-six-to-one, versus 18.7–22.4% for the other modes. This motivates retaining mode-specific diagnostics alongside overall MAE.

Table 12 retains the broader all-six-versus-target-only comparison for four architectures. Separate target-only models have lower average MAE for all four, although individual modes can benefit when one model predicts all modes. These results concern the tested baseline configurations, not the potential value of multi-mode information in models designed explicitly around transport interactions.

Table 12: One model for all modes vs one model per mode for four spatio-temporal models at the district 24h horizon. Joint uses all six modes as inputs and outputs; Sep uses only the target mode as input and output. Bold marks the lower MAE per (model, mode). Δ is (Sep − Joint)/Joint.

	STID			AGCRN			STAEformer			MTGNN		
Mode	Joint	Sep	Δ	Joint	Sep	Δ	Joint	Sep	Δ	Joint	Sep	Δ
metro_bus	2.94	**1.79**	−39.0%	2.83	**2.05**	−27.8%	2.30	**1.81**	−21.0%	3.00	**1.86**	−38.1%
local_bus	19.36	**15.85**	−18.1%	21.71	**18.74**	−13.7%	16.51	**16.01**	−3.0%	20.80	**17.52**	−15.7%
subway	37.43	**30.98**	−17.2%	44.98	**41.17**	−8.5%	32.95	**29.37**	−10.9%	44.56	**34.14**	−23.4%
walk	33.14	**27.77**	−16.2%	53.02	**43.50**	−18.0%	**28.33**	29.80	+5.2%	37.87	**35.50**	−6.3%
car	35.32	**29.38**	−16.8%	37.96	**32.27**	−15.0%	31.12	**27.90**	−10.3%	37.13	**31.85**	−14.2%
other	19.58	**16.27**	−16.9%	21.72	**19.07**	−12.2%	16.98	**16.32**	−3.9%	**21.01**	22.60	+7.6%
Avg	24.63	**20.34**	**−17.4%**	30.37	**26.13**	**−14.0%**	21.37	**20.20**	**−5.4%**	27.40	**23.91**	**−12.7%**

Table 13: MAE by training-set sparsity, district 24h horizon. Bold marks the lowest reported MAE in each group. Evaluation follows the baseline windowing configurations: iTransformer, SOFTS, MTSMixer, and TimeMixer use 1,733 test windows; the remaining models use 1,709. HA evaluated on the former windows gives 0.031, 0.275, 1.337, and 34.458 for the four groups.

Model	Rare	Sparse	Medium	Dense
Evaluation cells	3.7%	13.2%	18.0%	65.0%
HA	**0.031**	0.276	1.339	34.475
Latest Value	0.032	0.292	1.760	213.258
STAEformer	0.220	0.543	1.508	**32.325**
STID	0.393	0.979	2.080	37.084
AGCRN	1.057	2.070	3.013	45.402
GWNet	0.659	1.340	2.074	41.496
MTGNN	0.305	0.856	2.132	41.358
iTransformer	0.036	**0.189**	**1.210**	38.586
SOFTS	0.032	0.195	1.230	40.423
MTSMixer	0.875	0.911	1.715	48.433
TimeMixer	0.200	0.463	1.587	48.021

C.5 Forecasting errors by OD sparsity

We stratify district 24h test errors by each (OD pair, mode)'s fraction r of training hours with positive flow: rare ($0 < r \leq 1\%$), sparse ($1\% < r \leq 10\%$), medium ($10\% < r \leq 50\%$), and dense ($r > 50\%$). Pairs with no positive training observations account for 0.05% of evaluation cells and are excluded from these four groups. Tables 13 and 14 report eleven baselines, pooling errors across modes.

HA has the lowest reported MAE on rare pairs, iTransformer on sparse and medium pairs, and STAEformer on dense pairs. On rare pairs, positive-target MAE ranges from 3.28 to 4.14, whereas zero-target MAE ranges from 0.015 to 1.043. Predicting flow when the target is zero therefore accounts for much of the difference on rare pairs. Reporting both components exposes errors hidden by overall MAE.

D Dataset Documentation for SeoulMMOD

D.1 Provenance

SeoulMMOD is built from the Seoul Metropolitan Government's "Seoul Metropolitan Area Living Mobility Data," jointly produced with KT (formerly Korea Telecom). We downloaded ZIP-compressed CSV files from January 2023 through December 2025 from the Seoul Open Data Plaza and converted them into released parquet files and the benchmark format in Section 3.1.

Table 14: MAE conditioned on positive ($y > 0$) and zero ($y = 0$) targets within each training-set sparsity group. The evaluation settings match Table 13.

	Positive-target MAE				Zero-target MAE			
Model	Rare	Sparse	Medium	Dense	Rare	Sparse	Medium	Dense
HA	3.30	3.10	2.71	36.02	0.015	0.144	0.853	3.422
Latest Value	3.32	3.29	3.77	223.29	0.016	0.151	1.048	11.195
STAEformer	3.49	3.36	3.16	33.76	0.204	0.411	0.922	3.505
STID	3.59	3.60	3.73	38.71	0.377	0.856	1.497	4.278
AGCRN	3.89	4.14	4.44	47.42	1.043	1.973	2.506	4.692
GWNet	4.14	4.36	4.14	43.36	0.643	1.199	1.346	3.847
MTGNN	3.28	3.17	3.61	43.17	0.291	0.747	1.609	4.839
iTransformer	3.32	3.41	3.57	40.35	0.020	0.038	0.379	3.206
SOFTS	3.34	3.42	3.65	42.26	0.016	0.044	0.378	3.385
MTSMixer	3.53	3.21	3.15	50.65	0.862	0.804	1.210	3.874
TimeMixer	3.42	3.34	3.31	50.21	0.184	0.328	0.981	4.024

D.2 Composition

Each record contains the population-scaled number of trips from origin zone i to destination zone j during hour t by mode m. SeoulMMOD uses two Seoul zone systems, 25 districts and 426 sub-districts, and six intra-urban modes, metro bus, local bus, subway, walk, car, and other.

For 2024, the district representation contains $8{,}784 \times 625 \times 6 \approx 3.3 \times 10^7$ entries and the sub-district representation contains $8{,}784 \times 181{,}476 \times 6 \approx 9.6 \times 10^9$ entries. The complete dataset covers 2023, 2024, and 2025 at both spatial scales, with aligned calendar, rainfall, and POI signals described in Section 3.3. The released parquet files store observed OD records sparsely and do not include rows for absent OD-pair, mode, and hour combinations. When building benchmark tensors, these absent cells are filled with zeros, yielding the dense shapes above. Two sub-district boundary changes, a 2023-02 rename and a 2025-08 split, are reconciled in the OD-flow records to maintain stable zone identities.

SeoulMMOD contains aggregated administrative-zone OD records only. It contains no raw subscriber records, trajectories, or device identifiers and does not support individual-level tracking or re-identification.

Recommended splits are the chronological 70/10/20 split within 2024 (Section 4.1) and the cross-year setup that trains on 2024 and tests on 2025 (Section 4.5).

D.3 Upstream Generation

The public source documentation describes how KT LTE/5G signalling logs are cleaned, how stays are separated from movements using a 30-minute criterion, and how mode labels are assigned. Intercity modes are inferred from the origin–destination zone pair. Subway and bus trips are assigned by matching inferred paths to published transit networks. Remaining trips are classified as car or walk by speed and distance, and unmatched trips are assigned to "other."

The original source scales each KT subscriber's trips up to a population-level estimate. The scaling uses KT market share, LTE/5G subscriber share, phone-on ratio, and a gender-age adjustment against the national resident-registration population. SeoulMMOD uses the published OD flows as its flow target. These corrections reduce estimation error from relying on one carrier's subscriber base but cannot remove all residual error from phone-use behaviour, visitors, or under-represented subscriber groups.

D.4 Our Preprocessing

Four preprocessing steps convert the original source records into SeoulMMOD.

1. *Spatial filtering.* Keep only Seoul-to-Seoul OD records.

2. *Mode filtering.* Drop intercity modes and keep the six intra-urban modes.
3. *Code reconciliation.* Map the 2023-02 rename and 2025-08 split to stable zone identities.
4. *Temporal unification.* Drop sub-hour timestamps and re-aggregate records to a 1 h grid.

Timestamps are aligned to Asia/Seoul, which has no daylight-saving discontinuities. Calendar covariates are normalised to $[0, 1)$. The project repository provides data builders for converting the released parquet files into benchmark tensors.

D.5 Uses and Limitations

SeoulMMOD is intended for multi-horizon OD forecasting, comparing joint training with one model per mode, cross-year generalisation, and context-aware modelling with the bundled rainfall, calendar, and POI signals. Individual-level tracking or re-identification is out of scope.

SeoulMMOD is a single-city benchmark. Strong performance here does not by itself establish transfer to other cities, operators, or zoning systems. The OD flows are upstream mobile-network estimates rather than direct counts of all trips. Although the source applies population scaling, estimation error may remain for visitors, phone-use patterns, or underrepresented groups.

D.6 Distribution

During review, the anonymised dataset and code are mirrored at `https://anonymous.4open.science/r/SeoulMMOD`. The public dataset will be hosted on Kaggle. The underlying Seoul data are distributed under the Korea Open Government License (KOGL) Type 1. The processed SeoulMMOD release is distributed under CC BY-NC 4.0, with attribution to the original Seoul Open Data Plaza source.

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